SECSR2043 OPERATING SYSTEMS
[20 Marks]

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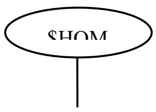
Instruction: Please answer all of the following questions. Whenever the ☝ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **$HOME** directory.

[4 marks]



Print screen the script that you type;

```
  GNU nano 7.2
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord"> accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date >dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .







afifshagir@secr2043:~$ chmod +x afif.sh~~_
```

Then draw the tree

```
    └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── fruit
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── greeting.txt
├── hello.c
├── hello.txt
├── hello_bak.txt
├── helloworld.jar
```

b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
  GNU nano 7.2
#!/bin/bash

read -p "Enter file extension (e.g., c,text):" ext

echo "DEBUG: Extension entered: '${ext}'"

temp_file=$(mktemp)
echo "DEBUG: Content of temporary file ($temp_file):"
cat "$temp_file"
echo "DEBUG: End of temporary file content."


find . -type f -name "*.${ext}">"$temp_file"

if [ ! -s "temp_file" ]; then
        echo "No '.${ext}' files found."
        rm "$temp_file"
        exit 0
fi

echo "Found files:"
cat "$temp_file"

num_files=%(wc -l < "<$temp_file")
echo "Total: $num_files"

rm "$temp_file"
```

```
afifshaqir@secr2043:~$ chmod +x short_finder.sh
afifshaqir@secr2043:~$ touch my_program.c
afifshaqir@secr2043:~$ touch another-code.c
afifshaqir@secr2043:~$ mkdir project
afifshaqir@secr2043:~$ touch project/utility.c
afifshaqir@secr2043:~$ touch document.txt
afifshaqir@secr2043:~$ touch script.sh
afifshaqir@secr2043:~$ ./short_finder.sh
Enter file extension (e.g., c,text):c
DEBUG: Extension entered: 'c'
DEBUG: Content of temporary file (/tmp/tmp.6Ds5SPVL0m):
DEBUG: End of temporary file content.
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

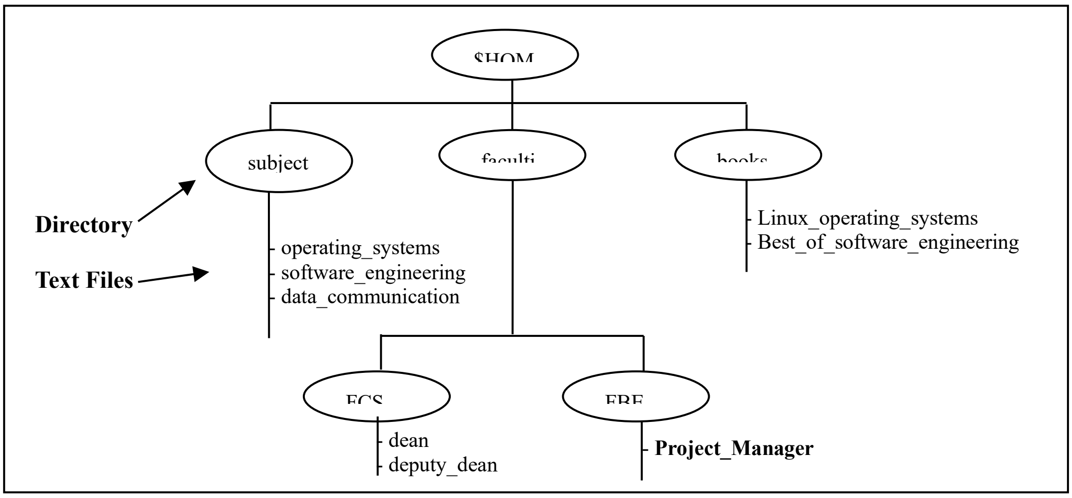
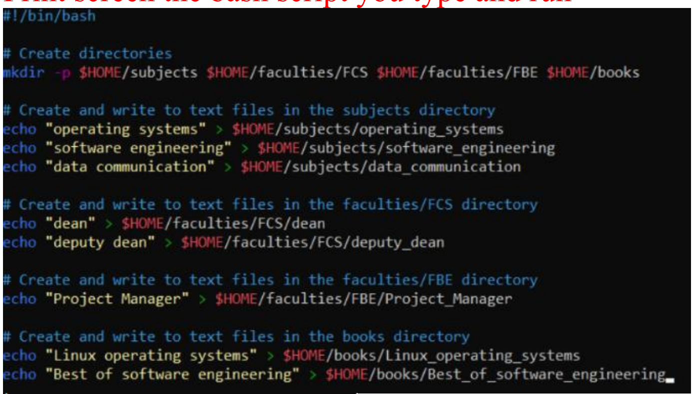


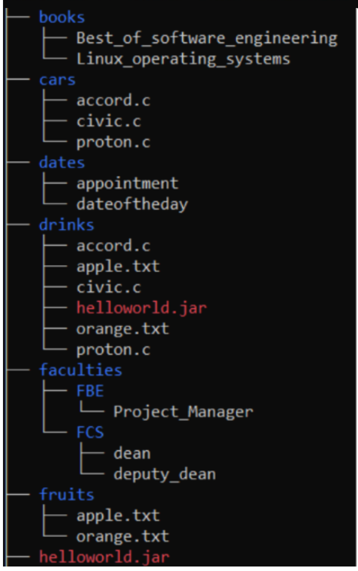
Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run



```
#!/bin/bash

# Create directories
mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

# Create and write to text files in the subjects directory
echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

# Create and write to text files in the faculties/FCS directory
echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

# Create and write to text files in the faculties/FBE directory
echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

# Create and write to text files in the books directory
echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
```



```
├── books
│   ├── Best_of_software_engineering
│   └── Linux_operating_systems
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── dates
│   ├── appointment
│   └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── faculties
│   ├── FBE
│   │   └── Project_Manager
│   └── FCS
│       ├── dean
│       └── deputy_dean
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── helloworld.jar
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxr-xr-x
Best_of_software_engineering	drwxr-xr-x
FCS	drwxr-xr-x
project_manager	drwxr-xr-x

```
drwxrwxr-x 2 afif afif 4096 Jun 16 2:09 books
drwxrwxr-x 2 afif afif 4096 Jun 16 2:09 subjects
drwxrwxr-x 2 afif afif 4096 Jun 16 2:09 FCS
```

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
#!/bin/bash

# Set permissions for subjects directory
chmod 744 $HOME/subjects

# Set permissions for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

# Set permissions for FCS directory
chmod 607 $HOME/faculties/FCS

# Set permissions for project_manager file
chmod 031 $HOME/faculties/FBE/Project_Manager
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr-xr-x
Best_of_software_engineering	drwxr-xr-x
FCS	drwxr-xr-x
project_manager	drwxr-xr-x

End of Lab 3

*** *All the Best for Final Exam* ***